02

ABSTRACT

Poor spinal posture is a growing health concern, particularly among students and professionals who spend long hours seated. This project presents a wearable smart posture monitor that uses a flex sensor and accelerometer to detect spinal bending in real time. When improper posture is detected, the system alerts the user via a connected Android application over Bluetooth. The goal is to build healthy postural habits and reduce long-term spinal problems.

1. INTRODUCTION

Back pain is one of the most widespread health issues today, affecting people across all age groups. A major contributing factor is prolonged poor posture during desk work or screen use. When the body becomes accustomed to a slouched position, muscles adapt to that alignment, making correction increasingly difficult over time.

This project addresses the problem by building a low-cost, wearable embedded system that continuously monitors the user's spinal posture. The system detects deviation from a healthy baseline and notifies the user to correct their position. Over time, consistent feedback trains the user to maintain good posture naturally.

2. COMPONENTS AND HARDWARE

2.1 ARDUINO UNO

The Arduino UNO serves as the central processing unit. Based on the ATmega328 microcontroller, it reads sensor inputs, processes the bending angle data, and triggers alerts when poor posture is detected. It features 14 digital I/O pins, 6 analog input pins, and operates at 16 MHz.

2.2 FLEX SENSOR

The flex sensor (SEN-08606, 4.5 inch) detects spinal bending by changing its electrical resistance as it bends. It is mounted along the back of the spine. Key specifications:

- Flat resistance: 10K ohm
- Bend resistance range: 60K – 110K ohm
- Operating voltage: 0–5V
- Power rating: 0.5W continuous, 1W peak

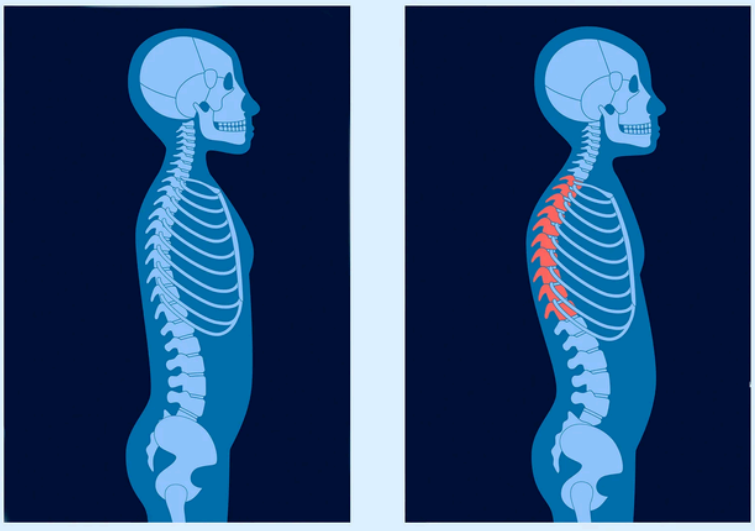
A voltage divider circuit with a 10K ohm resistor is used to read the output, which is fed into analog pin A4 of the Arduino.

2.3 ACCELEROMETER (ADXL335)

The ADXL335 accelerometer measures orientation and motion along three axes (X, Y, Z). This provides additional context about body position beyond simple bending. Its output pins (X_OUT, Y_OUT, Z_OUT) connect to analog pins A1, A2, and A3 on the Arduino respectively.

2.4 HC-05 BLUETOOTH MODULE

The HC-05 module enables wireless communication between the Arduino and the user's Android smartphone. It operates at 5V and connects via digital pins 4 (TXD) and 5 (RXD) on the Arduino. It can function in either master or slave configuration.



Healthy Spine Kyphosis

05

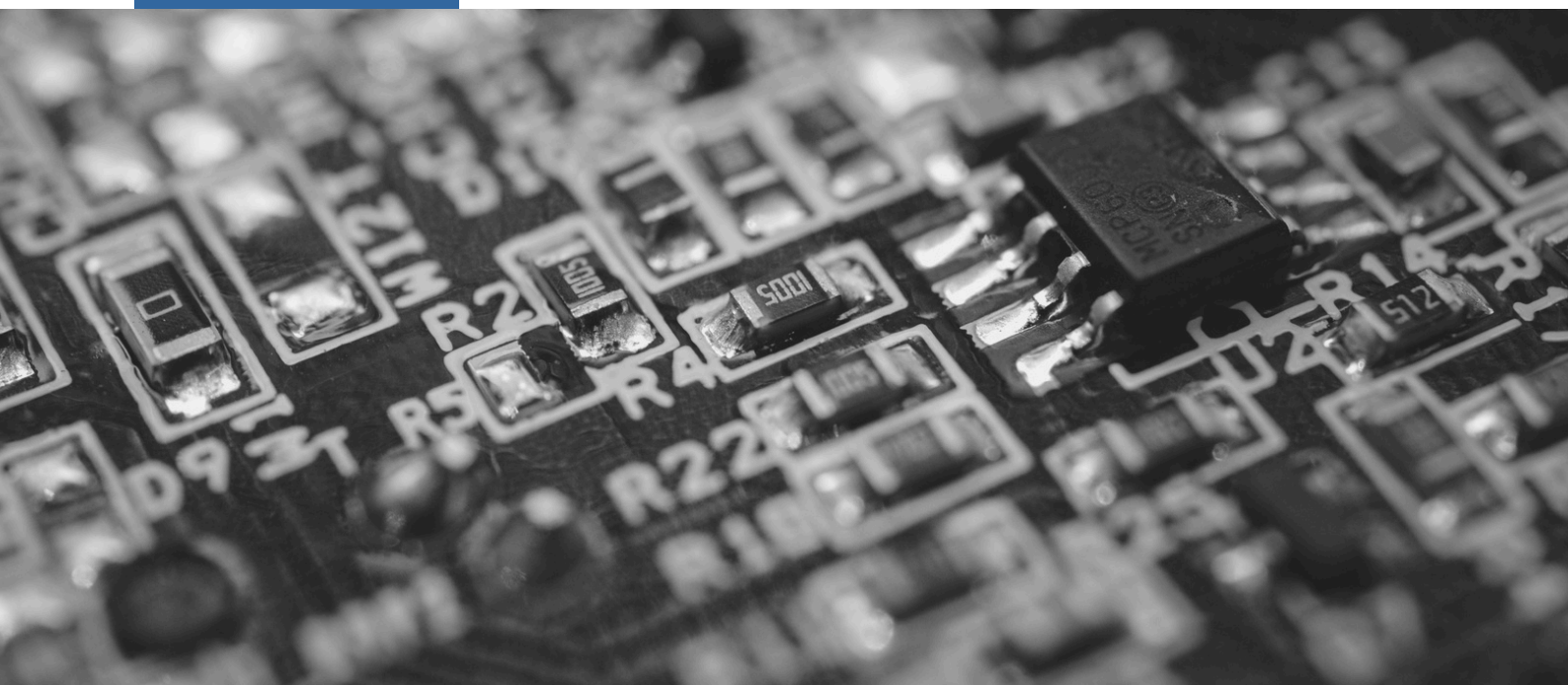
3. CIRCUIT CONNECTIONS

• Flex sensor pin 1 → 5V supply; pin 2 → Arduino analog pin A4 and one leg of 10K resistor

• Other leg of 10K resistor → GND

• ADXL335 VCC → 5V; GND → GND; X/Y/Z outputs → A1, A2, A3

• HC-05 VCC → 5V; RXD → digital pin 4; TXD → digital pin 5



4. SOFTWARE



4.1 ARDUINO FIRMWARE

The Arduino sketch reads analog values from the flex sensor and accelerometer continuously. It compares the bending angle against a predefined threshold set for the user. If the angle exceeds the threshold, the microcontroller sends a signal to the Bluetooth module to alert the user.

4.2 ANDROID APPLICATION

The Serial Bluetooth Terminal Android application is used to connect with the HC-05 Bluetooth module via Bluetooth to receive posture status data in real time. The app displays the incoming posture angle, surfaces visual text alerts for poor posture, and logs data history so the user can review their sitting patterns over time.

5. WORKING PRINCIPLE

The flex sensor and accelerometer are worn on the user's back. When the user sits upright, the sensor readings fall within the calibrated normal range. When the user slouches or bends forward excessively, the resistance of the flex sensor increases and the accelerometer detects a change in orientation. The Arduino processes these values and, if they fall outside the acceptable range, transmits an alert to the Android app via Bluetooth. The user is then prompted to correct their posture.

6. TESTING AND RESULTS

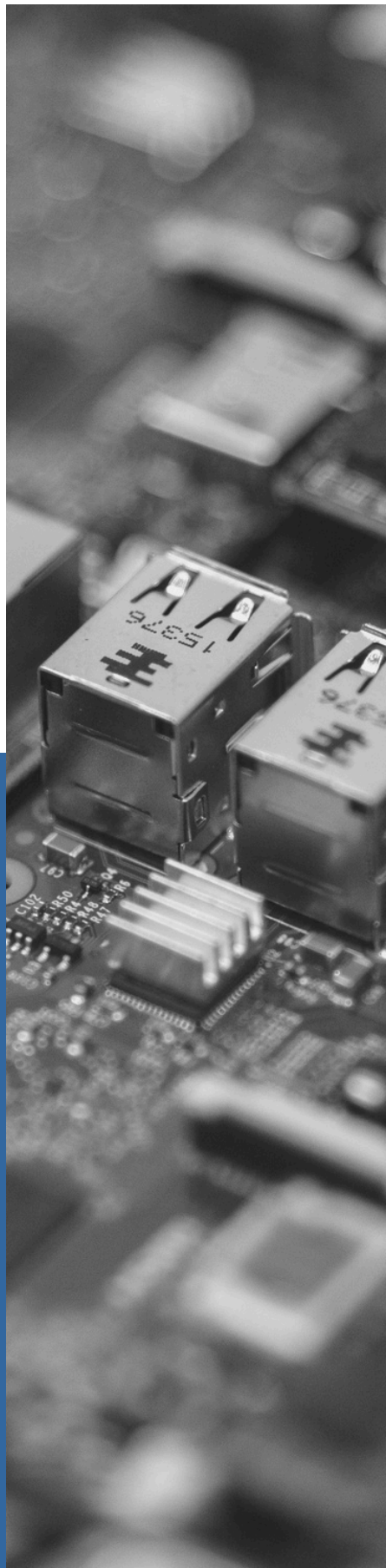
The system was evaluated across multiple distinct posture events to test its reliability and responsiveness. An angle of 0 degrees was established as the baseline for optimal, upright posture, while any measurement exceeding 300 degrees was defined as the threshold for poor posture. Throughout the various testing trials, the device consistently tracked angular transitions in real time, successfully triggering the alert system whenever a user breached the 300 degrees limit during slouching events.

7. CONCLUSIONS

The Smart Posture Monitor successfully demonstrates a practical embedded systems application for health monitoring. The system:

- Accurately detects improper spinal posture using a flex sensor and accelerometer
- Delivers real-time feedback to the user via a Bluetooth-connected Android app
- Encourages gradual improvement in postural habits through consistent monitoring

The project was implemented within hardware cost constraints and is suitable for static seated use.



8. FUTURE IMPROVEMENTS

- Extend the system to monitor dynamic activities such as walking or running
- Add wireless data logging to a PC or cloud platform for long-term analysis
- Miniaturize the hardware for a more comfortable wearable form factor
- Expand posture monitoring to cover the full body, not just the spine